

Hexadecimal & BCD · AS · Answers

U04 · 24 written + 6 pen-first = 30 · Student copy

Authored schemes, informed by published question shapes. Accept equivalent clear wording and valid working. Do not credit contradictions. Hex letter case and spaces between binary groups do not change a value.

1. Authored conversion table [4]

In order: **61**; **CA**; **1000 0100**; **E5**. One mark per complete value. BCD encodes 8 and 4 separately.

2. Hex to denary [2]

$12 \times 256 + 0 \times 16 + 9$ [1]; **3081** [1]. Accept equivalent valid working.

3. Denary to hex [2]

$427 \div 16 = 26$ r 11; $26 \div 16 = 1$ r 10; $1 \div 16 = 0$ r 1 [1]; **1AB** [1]. Equivalent place-value/binary working accepted.

4. BCD digits [4]

(a) **0100 0000 0110** [1]. (b) **281** [1]. (c) **1011** [1]; it denotes 11, outside the allowed decimal-digit range 0–9 [1]. The qualification is a decimal-digit nibble in 8421 BCD.

5. Same value, different encoding [3]

Ordinary binary **01001010** [1]; BCD **0111 0100** [1]. Binary represents the whole value using powers of 2, while BCD encodes decimal 7 then 4 separately [1].

6. Eight-bit value [1]

00001010 [1].

7. Applications [4]

(a) A suitable BCD application [1] plus its use explained [1], e.g. a decimal display system stores each digit as a nibble then uses a decoder/driver for the segment pattern. Accept a specific decimal-data application with an accurate explanation; do not assert that all banks/tills use BCD.

(b) A valid hex application [1] plus linked benefit [1], e.g. a debugger displays memory bytes in two-digit groups that are shorter to read and map directly to bits. Other accurate application/explanation pairs accepted.

8. Display claim [2]

A decoder/driver is still needed [1]; it maps the four-bit digit code to the on/off pattern of seven segments [1]. A BCD nibble does not directly specify seven independent segment states.

9. Compact notation [2]

Each hex digit corresponds uniquely to one four-bit pattern [1]; hence two digits preserve all eight bits while using fewer written symbols [1]. No claim of data compression or smaller underlying storage.

10. Pen-first translator [6]

(a) Group from the right and pad with zeroes on the left to a multiple of four [1]; translate each group using the nibble table and preserve order [1]. A clear numbered method or flowchart is acceptable. Equivalent correct algorithms accepted.

(b) **0001 1011 1001** [1]; **1B9** [1]. (c) **0000 1111** [1]; the supplied first hex digit is 0, which expands to four zero bits under the stated contract [1].

Unmarked RGB extension: channels **40**, **100**, **180**; new blue $196 = C4$, giving **#2864C4**. Red and green stay unchanged.

Optional coding challenge: no extra marks. Require both directions, correct left padding, four output bits per supplied hex digit, preserved leading zero nibbles, lowercase hex acceptance and rejection of invalid/empty input. The 13 local examples are not a proof. Check the `no-int()/hex()/bin()` rule by reading the converter functions. A nibble lookup table supplies a model approach.